

06 PROBLEM SOLUTION MAPPING

Problem–Solution Mapping

Traceability from each concrete disaster-management problem to a TRINETRA capability and demo signal.

Mapping principle

Every feature must trace to a specific user problem and a testable outcome. Features without a validated problem or demo value stay outside the MVP.

Specific problem	TRINETRA response	Evidence of working solution
Fragmented information	Central disaster dashboard	Active disasters, alerts, risk, resources, reports, and map context appear in one view.
Lack of localized information	Location-based map and risk context	A user can inspect relevant conditions and services for a selected area.
Delayed alerts	Centralized alert mechanism	An authority publishes a current, time-bound alert that citizens can see.
Difficulty finding safe locations	Safe-zone visualization	A citizen finds a nearby verified safe zone with useful details.
Difficulty finding emergency resources	Resource discovery	A citizen or responder filters nearby hospitals, shelters, and relief centers.
Citizen information does not reach authorities	Citizen reporting	A structured report reaches authority triage with a report ID and status.
Limited situational awareness	Authority dashboard + map + risk	Authorities can review incidents, reports, affected areas, alerts, and advisory risk together.

Validation questions

- Who experiences this problem, and in which disaster phase?
- What existing workaround do they use?
- Which information or decision is delayed?
- How will the team demonstrate improvement without inventing statistics?
- What verified source supports the problem claim?